

Broadband expansion raises small-firm productivity by 3.4 percent

Evidence from a staggered municipal fiber rollout

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Demonstration deck; setting and data are simulated.

Digital infrastructure could close the small-firm productivity gap

- Small firms employ most workers but modernize last (Syverson '11)
 - Fast internet could raise their productivity at low fiscal cost
- ⇒ Yet no causal estimate exists for the firms that need it most

Existing work has bypassed the small-firm segment

- Broadband's effect is measured in aggregate, not for small firms
 - Adoption is endogenous to a firm's own prospects
- ⇒ We exploit access timing that firms did not choose

Prior work: Czernich et al. '13; Akerman et al. '15; Hjort-Poulsen '19

This paper

1. First causal small-firm estimate: productivity rises **3.4%** (0.034)
2. Gains rise with communication intensity, as a simple model predicts
3. Back-of-the-envelope: the program repaid its cost within six years

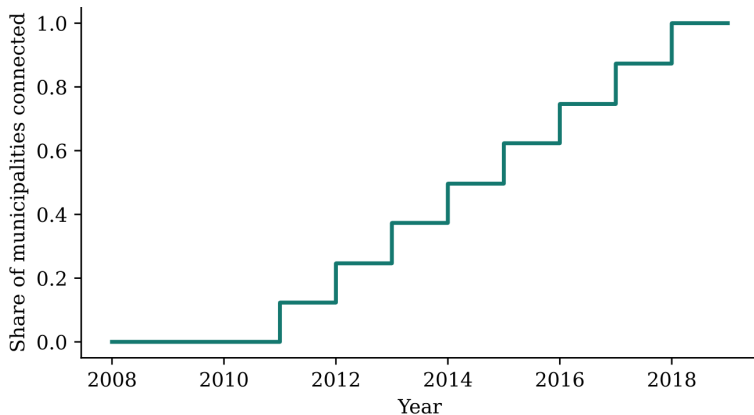
Implication: universal broadband is among the cheapest productivity policies for a small open economy.

Setting and data

- **Program:** all 284 municipalities wired to fiber, 2011–2018
- **Panel:** 9,400 firms with 3–49 employees, 2008–2019 (112,800 firm-years)
- **Outcome:** log real value added per worker
- **Intensity:** 2008 wage-bill share of communication tasks, fixed pre-program

Fiber reached towns one at a time, on a fixed schedule

Rollout order was fixed by the 2010 Act, before firms' outcomes were realized

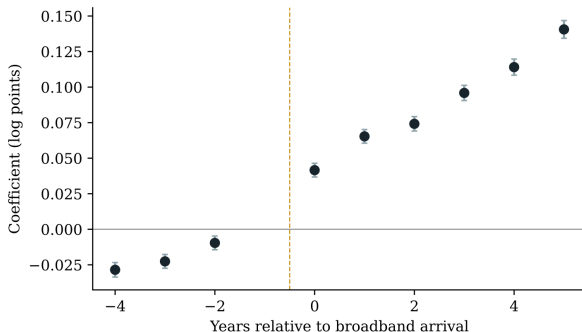


Scoring favored high-potential towns, yet a permuted-year placebo is a precise zero (Table 3).

▸ Placebo ▸ By cohort

Productivity is flat before arrival, then climbs steadily

Coefficients relative to the year before arrival; 95% CIs, firm-clustered



No pre-trends; the average post-arrival gain is 3.4% (Table 2, col 1).

► DiD table

► Robustness

Gains concentrate in communication-intensive firms

A task-outsourcing model predicts the effect rises with communication intensity (Prop. 1)

Dep. var.: log labor productivity	(1) Average	(2) + Interaction
Broadband	0.034*** (0.002)	0.020*** (0.003)
Broadband $\times$ comm. intensity		0.032*** (0.005)
Firm and year fixed effects	Yes	Yes
Observations	112,800	112,800
Interaction is positive and precise $\Rightarrow$ the model's complementarity holds.		

SEs clustered by firm. *** $p < 0.01$. Communication intensity is the 2008 wage-bill share.

A firm one SD above mean intensity gains about one-fifth more than the average firm.

The program paid for itself within six years

Policy return

A 3.4% productivity gain for the average connected small firm is worth about **410 million euros a year**. Against the fund's 2.3 billion euro construction cost, the small-firm channel alone repays the public investment within six years.

Scope: the mechanism is communication cost, not a Nordavian particularity, so the logic travels to other broadband programs.

Conclusion

1. Broadband arrival raises small-firm labor productivity by 3.4%
2. Gains concentrate where firms rely on communication-intensive tasks
3. The program repaid its cost within six years

Implication: universal broadband is among the cheapest productivity policies for a small open economy.

Appendix

Summary statistics

	Mean	SD	N
Log labor productivity	0.10	0.46	112,800
Employees	23.40	7.31	112,800
Firm age	21.45	9.56	112,800
Communication intensity	0.42	0.20	112,800
ICT investment (log)	0.37	0.36	112,800

Firm-year observations, 2008–2019. Balanced panel of 9,400 firms.

Model and estimating equation

A task-outsourcing model delivers the estimating equation

$$y_{it} - \ell_{it} = \alpha_i + \lambda_t + \underbrace{\beta B_{mt}}_{\text{average effect}} + \underbrace{\gamma (B_{mt} \times \theta_i)}_{\text{complementarity}} + \varepsilon_{it},$$

with B_{mt} a connected-municipality flag and θ_i communication intensity.

Proposition 1

The gain is strictly increasing in communication intensity: $G'(\theta_i) > 0$ on $[0, 1]$.

Baseline difference-in-differences estimates

Dep. var.: log labor productivity	(1) Baseline	(2) Firm controls	(3) Region trends	(4) Muni. cluster
Broadband	0.034*** (0.002)	0.033*** (0.002)	0.030*** (0.002)	0.034*** (0.003)
ICT investment (log)		0.000 (0.002)		
Firm and year fixed effects	Yes	Yes	Yes	Yes
Observations	112,800	112,800	112,800	112,800

SEs clustered by firm, except col. (4) by municipality. *** $p < 0.01$.

The 3.4% effect is stable across controls, trends, and clustering.

Robustness

Dep. var.: log labor productivity	(1) Region trends	(2) No controls	(3) Drop 2011–12 cohorts
Broadband	0.030*** (0.002)	0.034*** (0.002)	0.037*** (0.002)

All columns include firm and year fixed effects. SEs clustered by firm. *** $p < 0.01$.

Dropping the potentially non-random early cohorts nudges the estimate up, not down.

Placebo: permuted rollout years

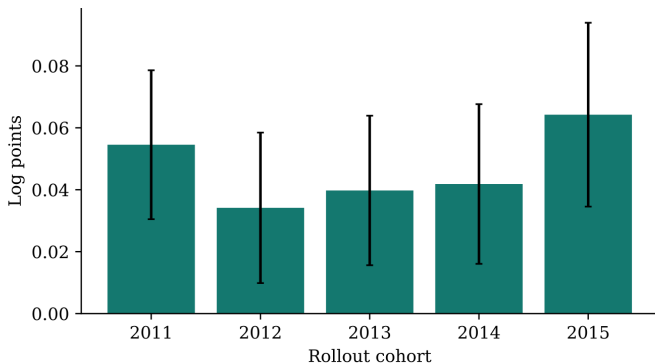
Dep. var.: log labor productivity	Pre-period placebo
Placebo broadband (permuted years)	−0.004 (0.002)

Pre-treatment sample. SEs clustered by firm.

A precisely estimated zero: the design recovers a program effect, not a trend.

Effects are similar across rollout cohorts

Each bar is a separate estimate vs. towns connected at least three years later; 95% CIs



Positive for every cohort, so negative weighting is not driving the result.

Related literature

- **Broadband:** Czernich et al. '13; Akerman et al. '15; Hjort–Poulsen '19
- **Staggered DiD:** Goodman–Bacon '21; Callaway–Sant'Anna '21; Sun–Abraham '21
- **Inference:** Bertrand et al. '04; Cameron–Miller '15; Abadie et al. '23